

Hallucination Detection in Large Language Models Using Multi-Signal Uncertainty Quantification

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A fully local, API-free framework integrating five complementary uncertainty signals into an ensemble for robust LLM hallucination detection.

Introduction & Problem Statement

Introduction

- **LLMs hallucinate:** outputs are grammatically fluent but factually incorrect, arising from statistical pattern matching rather than grounded knowledge.
- **High-stakes impact:** errors in healthcare, law, and education can cause serious harm—reliable detection is essential.
- **This work:** a modular, multi-signal detection system running entirely offline—no paid APIs, interpretable outputs, and ensemble-level robustness.

Problem Statement

Given a query Q and a generated response R :

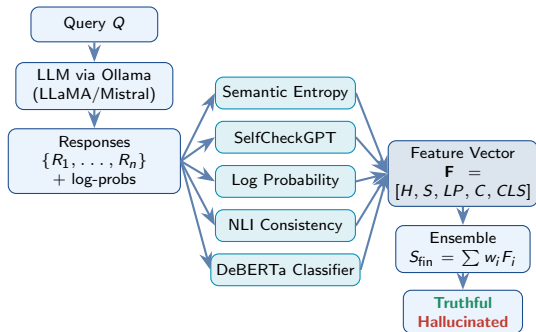
$$f(Q, R) \rightarrow \{\text{Truthful, Hallucinated}\}$$

System Constraints

- ✓ Fully offline — no external APIs
- ✓ Interpretable per-method signals
- ✓ High accuracy at reasonable cost

System Architecture & Dataset

Three-Stage Pipeline



TruthfulQA Dataset

- **817** benchmark questions
- **38** domains (health, law, science ...)
- Each sample: question + correct answers + misleading answers
- Reformulated as binary classification: $(Q, R) \rightarrow \{0, 1\}$
- Train / Test: **80% / 20%**, stratified

Class distribution

Truthful $\approx 45\%$ Hal-
lucinated $\approx 55\%$

Detection Algorithms

1. Semantic Entropy

$$H = - \sum_{i=1}^k p_i \log p_i$$

High $H \Rightarrow$ diverse meanings $\Rightarrow$ likely hallucination.

2. SelfCheckGPT

$$S = \frac{\# \text{ Entailments}}{\text{Total Comparisons}}$$

Low consistency across n generations $\Rightarrow$ hallucination signal.

3. Log Probability

$$\text{Score} = \frac{1}{n} \sum_{i=1}^n \log P(t_i \mid t_{<i})$$

Low score $\Rightarrow$ uncertain generation.

4. NLI Contradiction

$$C = \frac{\text{Contradictory Pairs}}{\text{Total Pairs}}$$

Cross-response contradictions indicate factual instability.

5. DeBERTa Classifier

- Input: [CLS] Q [SEP] R [SEP]
- Encoder: pretrained DeBERTa-v3
- Loss: $\mathcal{L} = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$

Ensemble Decision

$$S_{\text{final}} = \sum_{i=1}^5 w_i S_i; \quad \text{predict Hallucinated iff } S_{\text{final}} > \tau$$

Each method covers a *different failure mode* — combining them reduces both bias and variance.

Results & Evaluation

Per-Method Performance

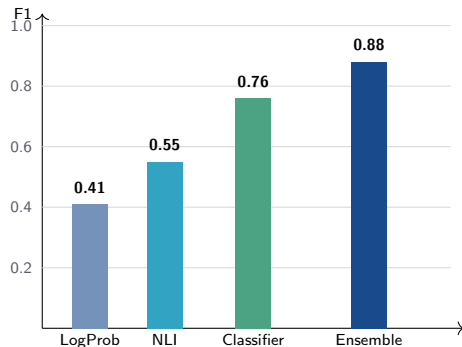
Method	Acc.	F1	Calls
Log Probability	0.54	0.41	1
NLI Consistency	0.48	0.55	2
Semantic Entropy	0.80	0.81	<i>n</i>
SelfCheckGPT	0.75	0.76	<i>n</i>
DeBERTa Classifier	0.76	0.76	0
Ensemble	0.87	0.88	all

Classifier (DeBERTa)

Class	Precision	Recall
Truthful	0.78	0.72
Hallucinated	0.74	0.80

ROC-AUC: **0.81**

F1 Score Comparison



Key insight: No single method is sufficient. Ensemble lifts F1 by +0.12 over the best individual method while maintaining interpretability.

Conclusion & Future Scope

Conclusion

- Hallucination is **multi-dimensional** — no single signal reliably captures it.
- Five complementary methods (semantic entropy, SelfCheckGPT, log-probability, NLI, DeBERTa) cover distinct failure modes.
- Ensemble achieves **F1 = 0.88**, **Acc. = 87%** on TruthfulQA; DeBERTa alone reaches ROC-AUC = 0.81.
- System is **fully local**, modular, interpretable, and deployable without paid APIs.

Future Scope

- **RAG Integration** — ground responses in retrieved knowledge to reduce hallucination at the source.
- **Real-time Optimization** — lightweight distilled models and parallel inference to cut latency.
- **Edge Deployment** — quantized models for on-device use (smartphones, embedded systems).
- **Adaptive Ensembles** — dynamically weight methods based on domain and query type.